

Track-level Train, Validation and Test Split Strategy

A reproducible multi-label stratification report for the 7,324-track MTG-Jamendo project subset

7,324	5,127	1,099	1,098
Total tracks	Training	Validation	Test

Executive summary

The dataset was divided at track level using iterative multi-label stratification over six target genres. A fixed seed of 42 makes the assignment reproducible. The first stage separated 5,127 training tracks from a 2,197-track holdout; the second stage divided that holdout into 1,099 validation and 1,098 test tracks. A deterministic correction moved the least disruptive records when the stratifier returned slightly different integer sizes. The final output contains exactly two columns: **track_id** and **split**.

All integrity checks passed: 7,324 unique assignments, no overlap between folds, complete coverage of source track IDs, exact requested fold sizes, and identical assignments when the full algorithm was rerun. Genre balance was strong: the maximum absolute difference between a fold prevalence and the full-dataset prevalence was 0.001996, or approximately 0.20 percentage points.

Scope decision

For these initial experiments, artist and album grouping were intentionally excluded. Consequently, this split measures performance under track-level randomization with label balancing; it does not measure strict generalization to unseen artists or albums. All windows belonging to one track must nevertheless remain in the same fold to prevent direct window-level leakage.

1. Inputs and split contract

The authoritative metadata table supplied exact track identifiers, while the genre table supplied six binary multi-label targets. Track identifiers were normalized only for joining, including removal of an optional track_ prefix and leading-zero differences; the final CSV preserved the exact identifiers from the metadata.

Contract item	Implemented rule
Population	7,324 unique tracks
Split unit	Whole track; never an individual 15-second log-Mel window
Target labels	classical, electronic, folk, hiphop, jazz, rock
Stratification	Iterative multi-label stratified shuffle splitting
Randomness	Seed 42 for stage 1; seed 43 for stage 2
Grouping	No artist or album constraint in this experimental version
Output	track_split_assignments.csv with track_id and split

Dataset genre support

The targets are multi-label, so genre-positive totals exceed the number of tracks. Every track had at least one target label, all target cells were binary, and the genre table agreed with the metadata labels.

Genre	Positive tracks	Full prevalence
Classical	1,446	19.74%
Electronic	1,733	23.66%
Folk	1,323	18.06%
Hiphop	1,304	17.80%
Jazz	1,376	18.79%
Rock	1,486	20.29%

2. Two-stage stratification procedure

Stage 1 - training versus holdout

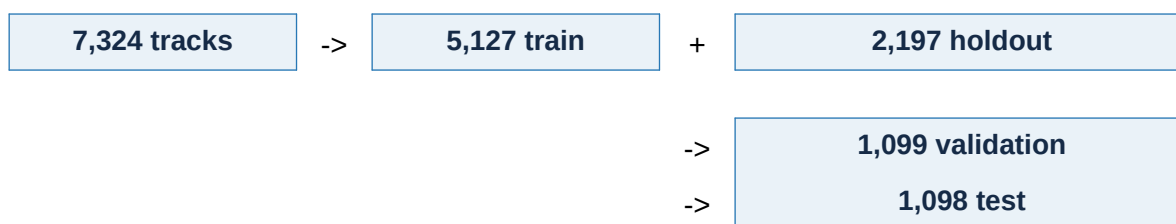
The first iterative split targeted a 2,197-track holdout. The library initially returned 5,141 training and 2,183 holdout tracks. The exact-size correction transferred 14 tracks from training to holdout, selecting at each step the candidate that minimized normalized squared error from the desired holdout genre-positive counts. The corrected result was 5,127 training and 2,197 holdout tracks.

Stage 2 - validation versus test

The 2,197-track holdout was then stratified independently. The initial result was 1,102 validation and 1,095 test tracks. The same deterministic correction moved three minimally disruptive tracks into test, producing 1,099 validation and 1,098 test tracks.

Why an exact-size correction was necessary

Iterative multi-label stratification prioritizes simultaneous preservation of multiple label distributions and may not honor an integer partition size exactly. The correction preserves the stratifier's solution as much as possible: only the size difference is moved, candidates are scored against expected positive counts, and seeded tie-breaking makes the operation reproducible.



3. Final fold balance

The table reports positive-label counts observed in the executed notebook. Close proportional agreement across folds confirms that the two-stage procedure retained the six-label distribution.

Split	Tracks	Classical	Electronic	Folk	Hiphop	Jazz	Rock
Train	5,127	1,010	1,209	924	911	961	1,038
Validation	1,099	218	262	199	196	207	224
Test	1,098	218	262	200	197	208	224

Genre prevalence by fold

Split	Classical	Electronic	Folk	Hiphop	Jazz	Rock
Full dataset	19.74%	23.66%	18.06%	17.80%	18.79%	20.29%
Train	19.700%	23.581%	18.022%	17.769%	18.744%	20.246%
Validation	19.836%	23.840%	18.107%	17.834%	18.835%	20.382%
Test	19.854%	23.862%	18.215%	17.942%	18.944%	20.401%

Maximum absolute prevalence deviation: 0.001996 (about 0.20 percentage points)

4. Validation, reproducibility and usage

Integrity checks passed

Check	Result
Total assignment rows	PASS - 7,324
Unique track IDs	PASS - 7,324
Missing split values	PASS - none
Exact fold sizes	PASS - 5,127 / 1,099 / 1,098
Train-validation overlap	PASS - none
Train-test overlap	PASS - none
Validation-test overlap	PASS - none
Coverage of source metadata	PASS - complete
Recreated assignment differences	PASS - 0
Genre prevalence tolerance	PASS - maximum 0.20 percentage points

How the split must be used

Join every branch table and log-Mel manifest to `track_split_assignments.csv` by normalized track ID. Every log-Mel window from a track inherits that track's fold. Fit Mel normalization, timbre and rhythm scalers, class weights, feature selection, and any learned preprocessing only on the training fold. Use validation for model selection and threshold tuning. Reserve test for the final locked evaluation.

Interpretation limitation

Because artist and album grouping were deliberately omitted, the same artist or album may appear in multiple folds. Results should therefore be described as track-level stratified performance, not unseen-artist or unseen-album generalization. A later robustness experiment should construct an artist-disjoint split and compare the performance gap.

Reproduction record

Item	Value
Notebook	02_create_final_track_splits_processed.ipynb
Method	MultilabelStratifiedShuffleSplit plus deterministic exact-size correction
Seeds	42 (train/holdout), 43 (validation/test)
Output	track_split_assignments.csv
Columns	track_id, split